

Preventing Representation Collapse in Joint-Embedding Predictive Architectures (JEPAs) for Edge Devices

The role of regularizer placement, centralized and decentralized

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Declaration

It is hereby declared that this thesis is my own original work while completing my degree at BRAC University. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.

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Approval

The thesis titled “Preventing Representation Collapse in Joint-Embedding Predictive Architectures (JEPAs) for Edge Devices: The Role of Regularizer Placement, Centralized and Decentralized” submitted by Atif Karim (Student ID: 1000055796) has been accepted as satisfactory in partial fulfillment of the requirements for the degree of Master of Science in Computer Science and Engineering.

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Abstract

As artificial intelligence moves toward decentralized deployment across local edge and Dew computing nodes, preserving representational quality without relying on cloud-based computation remains a critical challenge. Centralized architectures impose communication latency and data-privacy risks, whereas conventional lightweight models are prone to representation degradation—and, in the non-generative case, to latent collapse—when trained on resource-constrained micro-architectures.

This thesis introduces **EdgeVerify**, a Joint-Embedding Predictive Architecture (JEPA) designed for localized execution across the Dew and Edge computing tiers. Rather than reconstructing high-dimensional token sequences or raw pixels, EdgeVerify predicts context representations directly within an abstract joint-embedding space, using an Exponential-Moving-Average (EMA) target encoder and an action-conditioned latent predictor. To prevent dimensional collapse during localized, layer-wise gradient updates, we integrate VICReg-inspired variance and covariance regularization directly into the distillation objective, removing the need for negative-sample pairing.

We present a proof-of-concept implementation of the architecture and loss formulation, together with a reproducible experimental protocol for measuring convergence stability, representation collapse (embedding standard deviation and effective rank), and inference cost. A proof-of-concept evaluation against a non-regularized baseline—conducted on structured, digits, and CIFAR-10 data, and reproduced across two independent environments—shows that the variance constraint reliably prevents variance collapse (sustaining a mean embedding standard deviation above the hinge threshold), while revealing that at the current covariance weight the objective does not prevent dimensional (rank) collapse; this pattern holds on the CIFAR-10 natural-image subset as well as the smaller datasets. A systematic placement study isolates the cause: applying the variance penalty to an embedding *without* a co-located covariance penalty on the same embedding induces rank collapse that is worse than no regularization and invisible to both the training loss and the variance criterion. The implied correction—co-locating both penalties on the context embedding—restores effective rank by more than an order of magnitude on all three datasets and recovers linear-probe accuracy to the unregularized level (digits $0.56 \rightarrow 0.94$; CIFAR-10 $0.31 \rightarrow 0.41$). In a federated (FedAvg) setting the correction’s collapse resistance transfers, and it holds under non-IID label skew with up to ten clients, whereas the flawed placement collapses in every regime. Critically, however, a larger-scale study (ResNet-18 on full CIFAR-10) shows the extreme collapse to be largely a small-model artifact and—more consequentially—that effective rank ceases to predict downstream accuracy at scale: the highest-rank configuration is the *least* useful, and no regularizer beats

an unregularized baseline. The rank–usefulness correspondence, and the practical benefit of the reformulation, are therefore *scale-dependent*. The architecture itself is compact (~ 0.75 M parameters, 2.84 MB); characterizing the scale behavior with longer, multi-seed training, along with on-device measurement, is the primary item of future work.

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Chapter 1

Introduction

1.1 Background

The proliferation of deep learning models has accelerated interest in executing intelligent workloads directly on network boundaries and personal devices. Traditional cloud-centric AI architectures require the constant transmission of sensitive telemetry to remote servers, creating bandwidth bottlenecks, latency overheads, and substantial privacy risks [1]. Edge and Dew computing paradigms respond to these pressures by relocating inference—and, increasingly, learning—toward the device tier, where computation occurs close to, or directly at, the data source [2].

A parallel shift has occurred in the design of representation-learning objectives. Generative and reconstructive models expend significant capacity predicting high-dimensional outputs (pixels or tokens), much of which is irrelevant to downstream tasks. Joint-Embedding Predictive Architectures (JEPAs) instead predict in an abstract latent space, discarding unpredictable low-level detail and concentrating capacity on semantically meaningful structure [3], [4]. This property makes JEPAs an attractive candidate for the edge, where compute and memory budgets preclude large generative backbones.

1.2 Problem Statement

Executing predictive architectures on localized edge devices introduces two distinct and coupled challenges:

1. **Resource Constraints:** Limited memory bandwidth and VRAM make end-to-end backpropagation across large backbones infeasible on micro-architectures, motivating localized, layer-wise training schemes that bound the active computational graph.
2. **Latent Collapse:** Non-generative architectures such as JEPAs can trivially minimize a latent prediction loss by mapping disparate inputs to a constant embedding vector, collapsing representational capacity. Avoiding this failure mode without negative-sample pairing requires explicit regularization [5].

1.3 Research Objectives

- Design a localized, joint-embedding predictive architecture (EdgeVerify) tailored to edge and Dew computing nodes.
- Implement local, layer-wise gradient isolation to enable training under strict memory budgets by bounding the backpropagation graph.
- Integrate variance and covariance regularization into the distillation objective to prevent dimensional collapse during on-device learning.
- Define a reproducible protocol to evaluate convergence stability, representation collapse, inference latency, and memory footprint against cloud-reliant and non-regularized baselines.

1.4 Contributions

The contributions of this thesis are:

- An architecture, **EdgeVerify**, combining an EMA target encoder, an action-conditioned latent predictor, and a VICReg-inspired anti-collapse objective for edge deployment (Chapter 3).
- A stable local loss formulation, $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{distill}} + \alpha \mathcal{L}_{\text{var}} + \beta \mathcal{L}_{\text{cov}}$, that regularizes representations without negative pairs.
- A proof-of-concept implementation in PyTorch and an evaluation methodology for assessing the framework on resource-constrained hardware (Chapter 4).

1.5 Scope and Delimitations

The wider vision that motivates this project—decentralized, multi-agent operation and hallucination mitigation at the edge—is broader than what a single thesis can deliver. To be precise about scope, that vision is treated as *motivation*, and the work here is deliberately narrower. Specifically: (i) the “multi-agent” aspect is studied as a minimal two-node federated (FedAvg) experiment (Section 4.5.8) rather than a full multi-agent system with many heterogeneous, asynchronously communicating agents. (ii) “Hallucination mitigation” motivates the use of a non-generative, collapse-regularized representation, but no generative or downstream hallucination benchmark is evaluated; the study measures representation-level properties (collapse indicators and a linear probe). (iii) “Decentralized / on-device” execution motivates the lightweight design, but all measurements are taken on commodity CPUs rather than on deployed edge hardware. What the thesis contributes and evaluates is the single-node, variance-covariance-regularized joint-embedding core and an empirical characterization of its behavior. The broader claims are revisited as future work in Chapter 5.

1.6 Thesis Organization

Chapter 2 surveys related work in self-supervised representation learning, joint-embedding architectures, and edge/Dew computing. Chapter 3 details the EdgeVerify architecture, loss formulation, and training procedure. Chapter 4 presents the experimental protocol and reports results. Chapter 5 summarizes the findings, discusses limitations, and outlines future work.

Chapter 2

Literature Review

This chapter situates EdgeVerify within three bodies of work: (i) self-supervised representation learning and the problem of representational collapse, (ii) joint-embedding predictive architectures, and (iii) edge, Dew, and decentralized computing. For each, we identify the mechanisms relevant to the proposed design and the gaps that motivate it.

2.1 Self-Supervised Representation Learning

Self-supervised learning (SSL) removes the dependence on labeled data by constructing a pretext task from the input itself. Contrastive methods such as SimCLR [6] learn invariances by pulling together embeddings of augmented views of the same image while pushing apart embeddings of different images. Their principal limitation is a reliance on large batches of negative samples to avoid trivial solutions—a requirement that is poorly suited to memory-constrained edge hardware.

A subsequent line of work removed negatives entirely. BYOL [7] trains an online network to predict the output of a momentum-updated target network, demonstrating that a predictor combined with a stop-gradient and an Exponential Moving Average (EMA) target can prevent collapse without contrastive pairs. SimSiam [8] further showed that the stop-gradient operation, rather than the momentum encoder itself, is the critical component preventing collapse. EdgeVerify adopts this EMA-target-plus-predictor design (Chapter 3), but pairs it with explicit statistical regularization rather than relying on architectural asymmetry alone.

2.2 Regularization Against Dimensional Collapse

An alternative to asymmetric, negative-free designs is to regularize the statistics of the embedding space directly. Barlow Twins [9] minimizes redundancy by driving the cross-correlation matrix between two views toward the identity. VICReg [5] decomposes the objective into three explicit terms: an *invariance* term (mean-squared error between views), a *variance* term that maintains the standard deviation of each embedding dimension above a hinge threshold, and a *covariance* term that decorrelates dimensions. This explicit, non-contrastive formulation is

directly applicable at the level of individual layers and small batches, which is why EdgeVerify’s loss adapts the variance and covariance terms of VICReg. The distinction in this work is that the regularizers are applied within a *local, layer-wise distillation* objective rather than a single global loss.

2.3 Analyses of Representation Collapse and Decorrelation

A parallel body of work analyses collapse itself. Jing et al. [10] give a linear-model account of *dimensional collapse* in contrastive SSL, showing that embeddings can occupy a low-dimensional subspace even when a global collapse is avoided. Hua et al. [11] identify feature decorrelation as the key mechanism and propose explicit decorrelation to counteract it. Most directly relevant to this thesis, two works study *where* decorrelation must act in non-contrastive (BYOL/SimSiam-style) architectures: Li et al. [12] and Liu et al. [13] connect the predictor and stop-gradient asymmetry to an implicit decorrelation and analyse the conditions under which the representation collapses.

This thesis is closest to that last group, and it is important to state the relationship precisely: prior work has *already established*, for Siamese non-contrastive methods, that the placement of the decorrelation mechanism governs collapse. The present work does not discover that phenomenon; it extends it along three axes that the prior analyses do not cover. First, the setting is a *predictive* JEPA with an action-conditioned predictor and an *explicit* VICReg-style variance-plus-covariance objective, rather than a Siamese network with implicit decorrelation. Second, we show that the failure is *invisible* to both the training loss and the VICReg variance criterion—a diagnostic observation not emphasised in that work—and validate representation quality with a downstream linear probe. Third, we test whether the correct placement *transfers to decentralized (federated) training*. The contribution is therefore an extension of an established collapse phenomenon to a new architecture and training regime, not a new phenomenon.

2.4 Joint-Embedding Predictive Architectures

LeCun’s position paper on autonomous machine intelligence [3] argues for prediction in representation space as a route to efficient world models, framing the JEPA as a core building block. I-JEPA [4] instantiated this for images: given a context block, a predictor estimates the representations of masked target blocks produced by a target encoder, with no pixel-level reconstruction. This yields strong representations while avoiding the computational cost of generative decoding, and—unlike reconstructive masked autoencoders [14]—never materializes the high-dimensional output. EdgeVerify inherits this predict-in-latent-space principle and extends it with an *action-conditioned* predictor: the prediction is conditioned on a spatial action (a bounding box), aligning the architecture with the conditional structure of world models in [3].

2.5 Knowledge Distillation

The teacher–student framing of EdgeVerify is related to knowledge distillation [15], in which a student network is trained to match the outputs of a teacher. In the SSL setting, DINO [16] recast self-distillation with no labels as an effective pretraining strategy for vision transformers, using a momentum teacher and centering/sharpening to avoid collapse. EdgeVerify’s EMA target encoder plays the role of a slowly-evolving teacher, but the anti-collapse mechanism is the VICReg-style variance/covariance penalty rather than centering and sharpening.

2.6 Edge, Dew, and Decentralized Computing

Edge computing relocates computation from centralized clouds toward the data source to reduce latency and bandwidth and to improve privacy [1]. Dew computing pushes this further, positioning independent, on-device capability that can function with minimal or intermittent connectivity to higher tiers [2]. Decentralized learning approaches such as federated learning [17] keep raw data local and communicate only model updates, mitigating privacy risk at the cost of communication rounds. EdgeVerify targets a stricter regime than federated learning: it aims to keep both data *and* training local to a single resource-constrained node, which sharpens the memory-budget problem and motivates the layer-wise gradient isolation described in the next chapter.

2.7 Summary and Research Gap

The reviewed literature establishes (i) that negative-free SSL is feasible via EMA targets and predictors, (ii) that explicit variance/covariance regularization prevents collapse without negatives, (iii) that JEPAs avoid the cost of generative decoding, and (iv) that, in Siamese non-contrastive methods, the placement of the decorrelation mechanism governs collapse [12], [13]. What remains underexplored—and what this thesis addresses—is (a) whether the same placement effect holds in an *explicitly* regularized predictive JEPA, (b) that the failure is invisible to the training loss and the variance criterion and must be detected by effective rank and a downstream probe, and (c) whether the corrected placement survives *decentralized* training on edge nodes. The thesis therefore extends an established collapse phenomenon to the JEPA and federated settings, using an action-conditioned architecture intended for on-device execution.

Chapter 3

Methodology

This chapter describes the EdgeVerify architecture, its training objective, and the localized training procedure. The formulation follows the proof-of-concept implementation exactly; parameter names and coefficients are preserved as defined in the reference code.

3.1 Architecture Design

EdgeVerify is a Joint-Embedding Predictive Architecture composed of four modules: a Context Encoder f_θ , an EMA-updated Target Encoder $f_{\bar{\theta}}$, an Action Encoder g_ϕ , and a Latent Predictor p_ψ . All embeddings share a common latent dimension $d = 256$.

3.1.1 Context and Target Encoders

The Context Encoder f_θ maps an input image $x \in \mathbb{R}^{3 \times H \times W}$ to a latent vector $s_t \in \mathbb{R}^d$ through three strided convolutional blocks (channel widths $32 \rightarrow 64 \rightarrow 256$, kernel size 3, stride 2), each followed by batch normalization and a ReLU nonlinearity, and terminated by adaptive average pooling and flattening:

$$s_t = f_\theta(x_{\text{partial}}), \quad s_t \in \mathbb{R}^d. \quad (3.1)$$

The Target Encoder $f_{\bar{\theta}}$ is architecturally identical and is initialized as a copy of f_θ . Its parameters are excluded from gradient computation and are instead updated by an Exponential Moving Average (EMA) of the context parameters:

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta, \quad \tau = 0.999, \quad (3.2)$$

where τ is the EMA decay. The target representation is computed under a stop-gradient:

$$s_{\text{target}} = \text{sg}[f_{\bar{\theta}}(x_{\text{full}})], \quad (3.3)$$

with $\text{sg}[\cdot]$ denoting the detached (no-gradient) operator. This asymmetry between a trainable online branch and a non-trainable EMA target is the mechanism, established by BYOL [7] and SimSiam [8], that prevents the two branches from collapsing to a shared constant.

3.1.2 Action-Conditioned Prediction

Prediction is conditioned on a spatial action $a \in \mathbb{R}^4$ (a normalized bounding box) encoded by g_ϕ , a two-layer multilayer perceptron ($4 \rightarrow 64 \rightarrow d$):

$$a_t = g_\phi(a), \quad a_t \in \mathbb{R}^d. \quad (3.4)$$

The Latent Predictor p_ψ ($2d \rightarrow 512 \rightarrow d$) takes the concatenation of the context and action embeddings and predicts the target representation:

$$s_{\text{pred}} = p_\psi([s_t \parallel a_t]), \quad s_{\text{pred}} \in \mathbb{R}^d. \quad (3.5)$$

Conditioning the predictor on an action aligns EdgeVerify with the world-model formulation of [3], in which future or masked representations are predicted as a function of a transformation applied to the context.

3.2 Loss Function and Regularization

To ensure representational stability without negative-sample pairing, the total objective combines a distillation term with the variance and covariance regularizers of VICReg [5]:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{distill}} + \alpha \mathcal{L}_{\text{var}} + \beta \mathcal{L}_{\text{cov}}, \quad \alpha = 1.0, \beta = 0.01. \quad (3.6)$$

Distillation term. $\mathcal{L}_{\text{distill}}$ is the mean-squared error between the predicted representation and the detached target:

$$\mathcal{L}_{\text{distill}} = \|s_{\text{pred}} - s_{\text{target}}\|_2^2 / d. \quad (3.7)$$

Variance term. The variance term enforces a per-dimension standard deviation above a hinge threshold $\gamma = 1.0$ on the context embedding s_t , preventing the informative branch from collapsing. Writing $\sigma_j = \sqrt{\text{Var}(s_{t,j}) + \epsilon}$ with $\epsilon = 10^{-4}$ over a batch of B instances,

$$\mathcal{L}_{\text{var}} = \frac{1}{d} \sum_{j=1}^d \max(0, \gamma - \sigma_j). \quad (3.8)$$

Covariance term. The covariance term decorrelates the dimensions of the predicted representation. Let $\tilde{s} = s_{\text{pred}} - \bar{s}_{\text{pred}}$ be the batch-centered prediction and $C = \tilde{s}^\top \tilde{s} / (B - 1)$ its covariance matrix; the term penalizes the off-diagonal energy:

$$\mathcal{L}_{\text{cov}} = \frac{1}{d} \sum_{i \neq j} C_{ij}^2. \quad (3.9)$$

Applying the variance penalty to the context embedding s_t and the covariance penalty to the prediction s_{pred} regularizes both the encoder output and the predictor output, discouraging both constant and correlated (rank-deficient) solutions.

3.3 Localized, Layer-Wise Gradient Isolation

To respect the memory budget of an edge node, EdgeVerify bounds the backpropagation graph rather than differentiating end-to-end through a large backbone. The stop-gradient on the target branch (Eq. 3.2) and the detachment of the teacher target act as a *firebreak*: gradients from $\mathcal{L}_{\text{distill}}$ do not propagate into $f_{\bar{\theta}}$, so only the online parameters $\{\theta, \phi, \psi\}$ retain an active computational graph at any step. This isolation is what makes multi-module training feasible under strict VRAM limits, and it is the property whose memory footprint Chapter 4 is designed to quantify.

3.4 Training Procedure

The model is optimized with AdamW (learning rate 10^{-3} , weight decay 10^{-4}). At each step, the online branch produces (s_{pred}, s_t) , the target branch produces s_{target} under stop-gradient, the total loss (Eq. 3.6) is backpropagated through the online parameters, and the target encoder is updated by the EMA rule (Eq. 3.2). Table 3.1 summarizes the configuration used by the proof-of-concept implementation.

Table 3.1: EdgeVerify configuration in the proof-of-concept implementation.

Component / Hyperparameter	Value
Latent dimension d	256
Encoder blocks (channels)	$32 \rightarrow 64 \rightarrow 256$
Action encoder	$4 \rightarrow 64 \rightarrow 256$
Predictor	$512 \rightarrow 512 \rightarrow 256$
EMA decay τ	0.999
Variance threshold γ	1.0
Variance weight α	1.0
Covariance weight β	0.01
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-4}

Note on the current data pipeline. In the present proof-of-concept, the training loop operates on synthetically generated tensors: full images are sampled from a uniform distribution, the partial (context) view is produced by zeroing a central 32×32 region, and a single fixed bounding box is used as the spatial action. This configuration verifies that the architecture trains and that the loss is numerically stable, but it does not constitute an evaluation on real data. Replacing this pipeline with a standard image dataset is a prerequisite for the empirical results discussed in Chapter 4, and is reflected in the experimental design below.

Chapter 4

Experimental Design and Results

Scope of this evaluation. The results below were produced by the accompanying `evaluate.py` on the three datasets described in Section 4.3: a procedurally generated *structured* image set, the offline scikit-learn *digits* set, and a subset of the standard *CIFAR-10* benchmark. The structured and digits runs were executed in the primary environment (a network-restricted cloud CPU) and independently reproduced on Google Colab; the CIFAR-10 run was performed on Colab, where the benchmark could be downloaded. All figures are genuine measurements of training dynamics and resource cost; no classification accuracy is claimed, as the evaluation is at the level of the self-supervised representation rather than a downstream classifier.

4.1 Research Questions

RQ1 (Collapse): Does the variance/covariance regularization prevent representational collapse relative to an unregularized ($\alpha = \beta = 0$) JEPA?

RQ2 (Efficiency): What are the inference latency and memory footprint of EdgeVerify?

RQ3 (Convergence): Does the local objective converge stably?

4.2 Experimental Setup

Experiments were run on CPU using PyTorch (version 2.14 in the primary cloud environment; the Colab reproduction used its default runtime). The architecture and optimizer follow Chapter 3 exactly (Table 3.1). The primary runs use a fixed seed (`seed=0`); a five-seed robustness study quantifying run-to-run variance is reported in Section 4.5.4. The structured and digits runs were additionally reproduced on an independent machine (Google Colab) to check reproducibility (Section 4.5.3). Table 4.1 lists the run configuration.

Table 4.1: Evaluation configuration.

Item	Value
Framework / device	PyTorch, CPU
Optimizer	AdamW (10^{-3} , weight decay 10^{-4})
Batch size	64
Epochs (structured / digits / CIFAR-10)	40 / 25 / 15
Samples (structured / digits / CIFAR-10)	768 / 1797 / 2000
Seeds	primary: 0; robustness study: {0, 1, 2, 3, 4}

4.3 Datasets

Structured. Each 64×64 RGB image is generated with a random background gradient and one to three random rectangles and circles, yielding genuine local spatial structure (in contrast to the i.i.d. noise used in the initial smoke test). The context view is produced by zeroing a random rectangular region occupying 35–60% of each side, and the spatial action is the normalized bounding box of that region.

Digits. The scikit-learn handwritten-digits dataset (1797 real 8×8 images) is upsampled to 64×64 and replicated to three channels. It provides a small but *real* data source available without network access, used as a sanity check. The same random masking is applied.

CIFAR-10. A subset of the standard CIFAR-10 benchmark (2000 natural 32×32 images, upsampled to 64×64) is used to test whether the findings on the smaller datasets transfer to real natural images. It was run in the Colab environment where the benchmark could be downloaded. The same random masking and spatial-action construction are applied. A subset and a reduced epoch budget are used for tractability on CPU; a full-dataset run is left to future work.

Figure 4.1 shows representative samples from the structured and digits datasets. The top row of each is the full image (the reconstruction target); the bottom row is the masked context view actually presented to the context encoder, together with the normalized bounding box $[x, y, w, h]$ supplied to the action encoder.

4.4 Evaluation Metrics

Two collapse indicators are computed over the context embedding $s_t \in \mathbb{R}^d$ ($d = 256$) on up to 512 samples:

- **Mean embedding std** $\bar{\sigma} = \frac{1}{d} \sum_j \sqrt{\text{Var}(s_{t,j})}$. Variance collapse drives $\bar{\sigma} \rightarrow 0$.
- **Effective rank** (participation ratio) $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, where λ_i are the eigenvalues of the embedding covariance. Dimensional collapse (correlated dimensions) drives this toward 1.

Efficiency is reported as parameter count, parameter memory, and single-sample inference latency (mean $\pm$ std over 50 runs after warm-up).

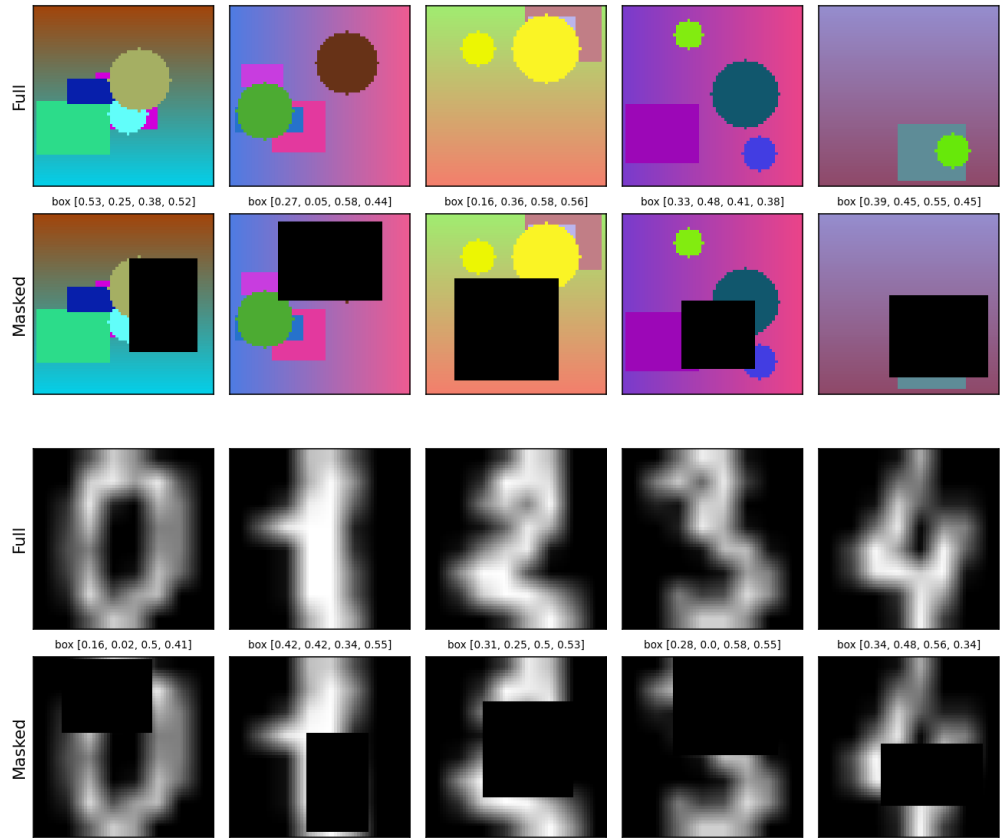


Figure 4.1: Representative training samples. Top: the procedurally generated *structured* dataset. Bottom: the real scikit-learn *digits* dataset (upsampled from 8×8). In each pair, the “Full” row is the target image and the “Masked” row is the context view with a random region zeroed; the box label is the spatial action.

4.5 Results

4.5.1 Convergence and Collapse (RQ1, RQ3)

Table 4.2 reports the final loss and collapse indicators; Figure 4.2 shows the training curves and Figure 4.3 the collapse indicators.

Table 4.2: Convergence and collapse indicators. $\mathcal{L}_{\text{total}}$ and $\mathcal{L}_{\text{distill}}$ are the final-epoch values; $\bar{\sigma}$ and effective rank are measured on the trained context encoder. A lower $\mathcal{L}_{\text{distill}}$ is not necessarily better—see the discussion.

Data	Model	$\mathcal{L}_{\text{total}}$	$\mathcal{L}_{\text{distill}}$	$\bar{\sigma}$	Eff. rank
Structured	Unregularized ($\alpha=\beta=0$)	0.0021	0.0021	0.245	6.46
	EdgeVerify (full)	0.190	0.120	1.549	1.64
Digits	Unregularized ($\alpha=\beta=0$)	0.0006	0.0006	0.080	3.57
	EdgeVerify (full)	0.257	0.207	1.772	1.03
CIFAR-10	Unregularized ($\alpha=\beta=0$)	0.0019	0.0019	0.223	4.53
	EdgeVerify (full)	0.231	—	1.684	1.36

The CIFAR-10 rows were obtained on Google Colab; the full-model $\mathcal{L}_{\text{distill}}$ component was not logged separately in that run and is marked “—”.

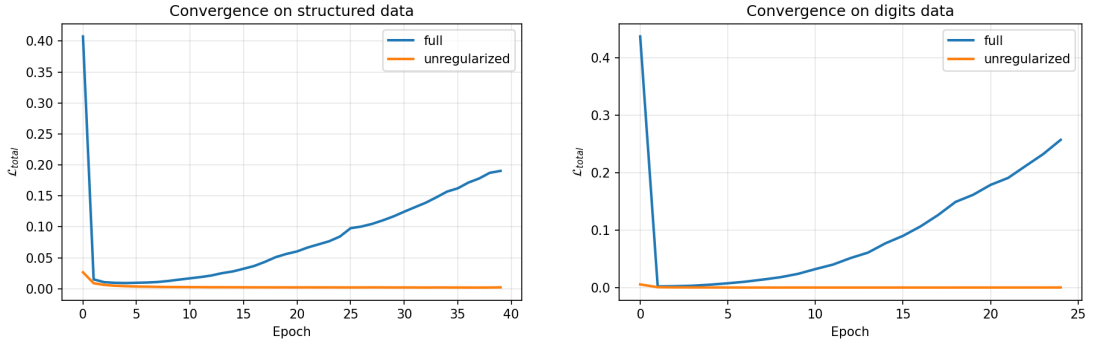


Figure 4.2: Total-loss convergence for the full objective and the unregularized baseline. The unregularized objective (distillation only) is trivially driven toward zero, whereas the full objective is non-monotonic: after the distillation term is minimized, the aggregate rises as the (weakly weighted) covariance term grows.

The results are mixed and warrant careful reading. The variance regularizer is effective at its stated objective: across all three datasets the full model maintains $\bar{\sigma} \approx 1.55$ – 1.77 (structured, digits) and 1.68 on CIFAR-10, at or above the hinge threshold $\gamma = 1.0$, whereas the corresponding unregularized baselines collapse toward low per-dimension variance ($\bar{\sigma} = 0.245, 0.080$, and 0.223). This directly confirms that the variance term prevents the low-variance mode of collapse.

However, the *effective rank* tells the opposite story: the full model attains a very low effective rank ($1.64, 1.03$, and 1.36 out of 256 for structured, digits, and CIFAR-10), and in every case it is *lower* than the corresponding unregularized baseline ($6.46, 3.57$, and 4.53). High per-dimension variance therefore coexists with

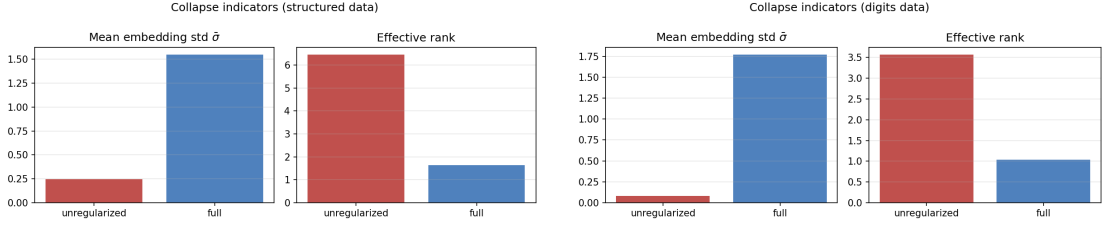


Figure 4.3: Collapse indicators. The variance term sustains a high mean embedding standard deviation $\bar{\sigma}$, but the effective rank of the full model is lower than that of the unregularized baseline.

strongly correlated dimensions—a rank-deficient representation. At the code’s covariance weight $\beta = 0.01$, the decorrelation term is too weak to counteract this, which is consistent with the growing covariance component visible in Figure 4.2. **The regularization as configured prevents variance collapse but does not, on this evidence, prevent dimensional (rank) collapse.** Crucially, this pattern is reproduced on the CIFAR-10 natural-image subset, indicating that it is not an artifact of the synthetic or low-resolution data.

4.5.2 Efficiency (RQ2)

Table 4.3 reports the model’s cost. The architecture is compact—745,536 parameters, 2.84 MB of parameter memory—and achieves single-sample inference in roughly 2 ms on CPU.

Table 4.3: Inference latency and memory. Peak RSS reflects the full PyTorch process (runtime libraries dominate) and is reported only as a coarse proxy; the model’s own parameter memory is the meaningful edge-relevant figure.

Data	Params	Mem. (MB)	Latency (ms)	Peak RSS (MB)
Structured	745,536	2.84	2.04 ± 0.11	852
Digits	745,536	2.84	2.48 ± 0.53	1099
CIFAR-10	745,536	2.84	2.58	—

The parameter count and memory are architecture-fixed and identical across datasets. Latency is CPU wall-clock and environment-dependent; the CIFAR-10 figure is from Colab (its per-run standard deviation and peak RSS were not recorded).

4.5.3 Independent Reproduction

The structured and digits experiments were re-executed from the same source on Google Colab (a distinct machine and software stack). The collapse indicators reproduced closely: the mechanism-level conclusions were identical, and the metrics matched within run-to-run variation (Table 4.4). The largest discrepancy was the digits full-model $\bar{\sigma}$ (1.77 original vs. 1.42 on Colab), while the effective-rank ordering—the finding of interest—was preserved in every case. This supports the reproducibility of the reported results.

Table 4.4: Reproduction check: original (cloud) vs. independent (Colab) runs. Effective-rank ordering (full < unregularized) is preserved throughout.

Data / Model	Metric	Original	Colab
Structured, full	$\bar{\sigma}$ / rank	1.549 / 1.64	1.551 / 1.61
Structured, unregularized	$\bar{\sigma}$ / rank	0.245 / 6.46	0.244 / 6.48
Digits, full	$\bar{\sigma}$ / rank	1.772 / 1.03	1.418 / 1.05
Digits, unregularized	$\bar{\sigma}$ / rank	0.080 / 3.57	0.079 / 3.65

4.5.4 Robustness Across Seeds

To quantify sensitivity to training stochasticity, the structured and digits experiments were repeated over five random seeds (data content held fixed; the seed varies weight initialization and minibatch order). Table 4.5 and Figure 4.4 report the mean and standard deviation of the collapse indicators. The separation between the full model and the unregularized baseline is consistent across seeds: on both datasets the error bars for effective rank do not overlap, and the direction of every comparison is preserved. The single-seed results reported above are therefore representative rather than incidental. The five-seed means were additionally reproduced on the independent Colab run (e.g. structured full effective rank 1.99 ± 0.25 vs. 1.99 ± 0.24 here).

Table 4.5: Collapse indicators over five seeds (mean $\pm$ std). The full-vs-unregularized ordering holds for every seed on both datasets.

Data	Model	$\bar{\sigma}$	Eff. rank
Structured	Unregularized	0.251 ± 0.005	6.58 ± 0.41
	EdgeVerify (full)	1.449 ± 0.068	1.99 ± 0.24
Digits	Unregularized	0.082 ± 0.004	3.30 ± 0.33
	EdgeVerify (full)	1.464 ± 0.237	1.05 ± 0.02

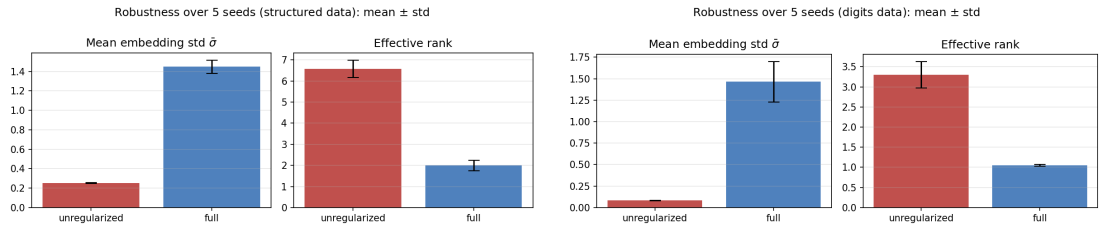


Figure 4.4: Collapse indicators over five seeds (mean $\pm$ std). Error bars for effective rank do not overlap between the two models on either dataset.

4.5.5 Covariance-Weight Ablation

The low effective rank of the full model raises the question of whether it is simply a consequence of an under-weighted covariance term. To test this, the covariance

weight β was swept over four orders of magnitude on the structured dataset with the variance weight fixed ($\alpha = 1.0$); Table 4.6 and Figure 4.5 report the outcome.

Table 4.6: Covariance-weight (β) ablation on structured data ($\alpha = 1.0$, single seed). Increasing β raises effective rank monotonically, but by a small margin.

β	$\bar{\sigma}$	Eff. rank
0.01 (default)	1.551	1.61
0.10	1.567	1.82
1.00	1.567	1.96
10.0	1.471	2.19

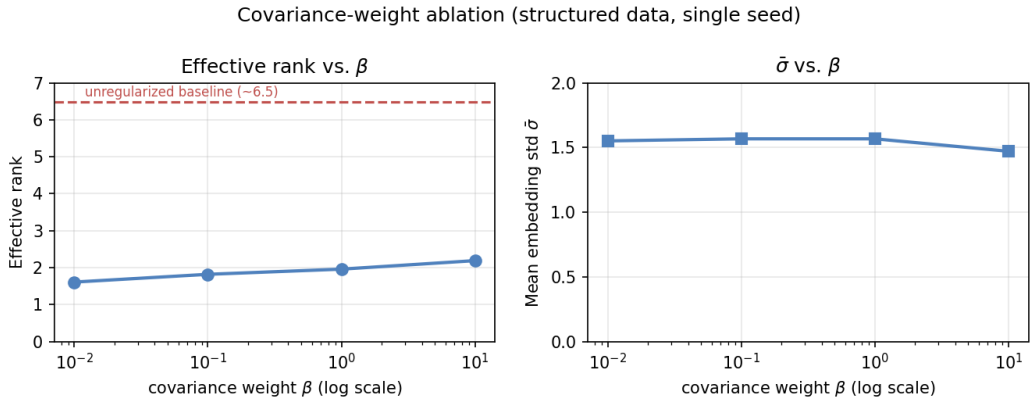


Figure 4.5: Covariance-weight ablation. Effective rank increases monotonically with β but remains far below the unregularized baseline (~ 6.5 , dashed) even at $\beta = 10$; mean embedding standard deviation is essentially unaffected.

The ablation is informative in a negative sense. Increasing β does move effective rank in the expected direction—monotonically, from 1.61 at the default to 2.19 at $\beta = 10$ (a $1000\times$ increase)—confirming that the covariance term is directionally effective. However, the magnitude is small: even at $\beta = 10$ the effective rank (2.19) remains far below both the unregularized baseline (~ 6.5) and the 256-dimensional ceiling, while $\bar{\sigma}$ stays essentially constant. Rank collapse is therefore *not* attributable merely to an under-weighted covariance penalty. A more likely cause is structural: the variance and covariance penalties are applied to *different* tensors (the context embedding s_t and the prediction s_{pred} ; Chapter 3), so the term that sustains variance and the term that decorrelates never act on the same representation. Testing a formulation that applies both penalties to a common embedding is identified as the principal next step (Chapter 5).

4.5.6 Regularizer-Placement Study

The β ablation indicated that collapse is governed by *where* the penalties are applied, not by their weight. We test this systematically. Each of the two penalties can act on either the context embedding s_t or the predictor output s_{pred} , giving a 2×2 placement grid, which we evaluate at fixed weights ($\alpha = \beta = 1.0$) on both

datasets. In all cases the diagnostics are computed on the frozen context encoder—the representation a downstream user consumes. Table 4.7 and Figure 4.6 report the grid.

Table 4.7: Regularizer-placement grid ($\alpha = \beta = 1.0$, single seed). “Var-loss” is the final variance term; a value of 0 means the variance criterion is fully satisfied. Only the placement *variance on s_t , covariance on s_{pred}* —the original wiring—collapses.

Data	Var. on	Cov. on	$\bar{\sigma}$	Eff. rank	Var-loss	Probe
Structured	s_t	s_{pred}	1.553	1.98	0.000	—
	s_t	s_t	0.311	47.44	0.691	—
	s_{pred}	s_{pred}	0.420	7.59	0.743	—
	s_{pred}	s_t	0.085	11.73	0.386	—
Digits	s_t	s_{pred}	1.365	1.02	0.000	0.637
	s_t	s_t	0.292	34.15	0.711	0.937
	s_{pred}	s_{pred}	0.346	9.62	0.749	0.920
	s_{pred}	s_t	0.057	4.71	0.479	0.933

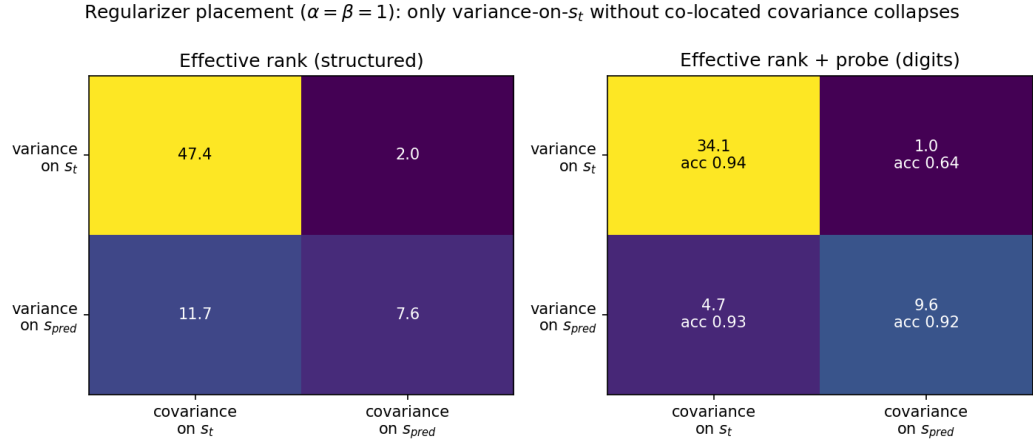


Figure 4.6: Effective rank across the placement grid. A single cell—variance on s_t with covariance on s_{pred} (top-right, the original wiring)—collapses on both datasets; every other placement preserves rank and (on digits) downstream accuracy.

Three observations follow, and together they constitute the central finding of this work. First, the collapse is *specific*: of the four placements, only one—variance on s_t with covariance on s_{pred} , which is exactly the original wiring—drives effective rank to ≈ 1 –2; every other placement yields rank ≥ 4.7 and, on digits, probe accuracy ≥ 0.92 . Second, the harmful ingredient is identifiable: applying the variance (scale) penalty to s_t *without* a co-located covariance (decorrelation) penalty on s_t inflates each dimension’s magnitude while leaving the dimensions free to correlate, producing a high-variance but rank-deficient representation. Placing the covariance penalty on s_t (whether or not the variance penalty is also there) avoids the collapse. Third, and most cautionary, the failure is *invisible to the objective*: in the collapsed cell the final variance loss is exactly 0.000—the VICReg variance

criterion is fully satisfied—and $\bar{\sigma}$ is the *largest* of any cell, yet the representation is rank-1 and least useful downstream. Neither the training loss nor the variance criterion signals the problem; only effective rank and the linear probe do.

We state the resulting design principle: *in negative-free predictive SSL, a variance penalty applied to an embedding must be paired with a covariance penalty on the same embedding; a variance penalty without co-located decorrelation is worse than no regularization at all.* This is not a single-seed artifact. Table 4.8 repeats the entire grid over five seeds: the ordering is preserved in every case, with small variance. The collapsed cell stays at rank ≈ 1 –2 while the healthy cells remain an order of magnitude higher, and the separations far exceed the standard deviations—the collapsed and healthy configurations do not overlap.

Table 4.8: Regularizer-placement grid over five seeds (mean $\pm$ std). The ordering is stable and the collapse of the (variance on s_t , covariance on s_{pred}) cell is not a single-seed effect.

Data	Var. on	Cov. on	Eff. rank	Probe
Structured	s_t	s_{pred}	2.19 ± 0.15	—
	s_t	s_t	45.94 ± 1.22	—
	s_{pred}	s_{pred}	7.75 ± 0.17	—
	s_{pred}	s_t	11.45 ± 0.16	—
Digits	s_t	s_{pred}	1.02 ± 0.01	0.618 ± 0.027
	s_t	s_t	34.53 ± 0.20	0.938 ± 0.007
	s_{pred}	s_{pred}	9.19 ± 1.52	0.907 ± 0.011
	s_{pred}	s_t	4.38 ± 0.43	0.929 ± 0.006

4.5.7 Shared-Embedding Reformulation and Downstream Validation

Guided by the placement study, we adopt the best cell—both penalties on s_t , the *shared-embedding* variant (`shared_cov=True`, $\alpha = \beta = 1.0$)—and validate it more thoroughly, extending the downstream linear probe to CIFAR-10. The probe freezes the trained context encoder, extracts features from the (unmasked) images, and fits a linear classifier on the class labels. Probe accuracy is a downstream, ground-truth measure of representation usefulness that the intrinsic collapse metrics lack. Structured data carries no labels, so the probe is reported for the labeled digits and CIFAR-10 sets.

Probe accuracy is 10-class (chance = 0.10). CIFAR-10 accuracies are low in absolute terms because the lightweight encoder, 2000-image subset, and 15-epoch budget form a deliberately small setup for natural images; the relevant signal is the *relative* ordering across configurations.

The result is instructive on three points. First, the structural diagnosis is confirmed on all three datasets: the shared-embedding variant raises effective rank from 1.64 to 47.44 (structured), 1.03 to 34.15 (digits), and 1.34 to 39.24 (CIFAR-10)—an improvement more than an order of magnitude larger than anything the β sweep achieved, and one that holds on real natural images. Second, the linear

Table 4.9: Shared-embedding reformulation. Effective rank is restored on all datasets; the linear probe (digits and CIFAR-10) measures the effect on *usefulness*. On both probes the original wiring’s rank collapse gives the lowest downstream accuracy, and the shared-embedding variant recovers it to the unregularized level.

Data	Model	$\bar{\sigma}$	Eff. rank	Probe acc.
Structured	Unregularized	0.245	6.46	—
	Original (full)	1.549	1.64	—
	Shared-embedding	0.311	47.44	—
Digits	Unregularized	0.080	3.57	0.944
	Original (full)	1.772	1.03	0.561
	Shared-embedding	0.292	34.15	0.937
CIFAR-10	Unregularized	0.226	4.59	0.415
	Original (full)	1.688	1.34	0.313
	Shared-embedding	0.288	39.24	0.405

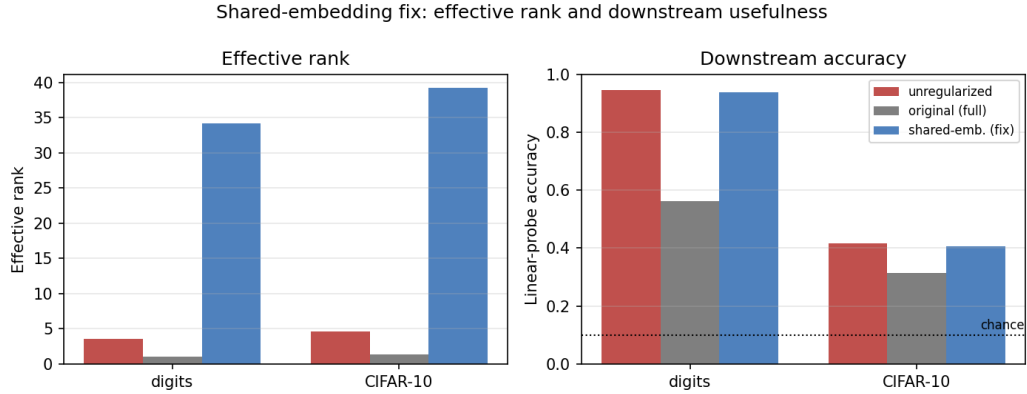


Figure 4.7: Effect of the shared-embedding reformulation on digits and CIFAR-10. Moving the covariance penalty onto s_t raises effective rank by more than an order of magnitude on both datasets. On the downstream linear probe, the original (full) configuration is worst on both, and the shared-embedding variant restores accuracy to the unregularized level. Note that the fix reaches *parity* with the unregularized baseline rather than exceeding it, and that absolute CIFAR-10 accuracy is low (see text).

probe shows the rank collapse was not benign: on *both* probes the original configuration is the worst (accuracy 0.561 on digits and 0.313 on CIFAR-10), below both the unregularized baseline (0.944; 0.415) and the shared-embedding variant (0.937; 0.405). In other words, the regularizer as originally wired *harmed* the representation relative to using no regularization at all, and the shared-embedding reformulation repairs this on both datasets. It is important to be precise about the magnitude of that repair: the fix restores accuracy to *parity* with the unregularized baseline, not above it, so the contribution is the removal of a harmful failure mode rather than a net gain over no regularization. Third, the probe validates the choice of diagnostic: the original model has the largest $\bar{\sigma}$ yet the lowest accuracy, whereas accuracy tracks effective rank—confirming that effective rank, not embedding standard deviation or loss magnitude, is the metric aligned with representation quality here.

Several caveats apply. These runs use a single seed. Absolute CIFAR-10 accuracy is low (~ 0.4) because of the deliberately small setup (lightweight encoder, 2000-image subset, 15 epochs), so the CIFAR-10 probe should be read as a relative comparison, not an accuracy benchmark. The shared-embedding variant also lowers $\bar{\sigma}$ (energy is spread across many more dimensions), so the variance hinge is no longer saturated; the probe indicates this does not harm usefulness, but a joint tuning of α and β under the shared formulation is a natural follow-up.

4.5.8 Decentralized (Multi-Agent) Training

The edge and Dew setting that motivates EdgeVerify is inherently decentralized, so a natural question is whether the placement finding and the shared-embedding fix survive when training is distributed across agents rather than centralized. We test this with a minimal two-node federated scheme. The data are split into two disjoint shards; each node trains locally on its shard for several epochs; after each communication round the nodes’ online parameters (context encoder, action encoder, and predictor, including BatchNorm statistics) are averaged (FedAvg) into a global model and broadcast back, and the target encoders are re-synchronised to the global context encoder. We run 6 rounds (local epochs 6/4 for structured/digits) and compare three settings: centralized shared-embedding training (reference), federated shared-embedding, and federated with the original wiring. Table 4.10 and Figure 4.8 report the outcome.

Table 4.10: Two-node FedAvg training (mean $\pm$ std over three seeds). The shared-embedding fix’s collapse resistance transfers to the decentralized setting; the original wiring collapses in both regimes.

Data	Setting	Eff. rank	Probe
Structured	Centralized (shared)	44.16 ± 1.22	—
	Federated (shared)	26.89 ± 0.74	—
	Federated (original)	2.69 ± 0.04	—
Digits	Centralized (shared)	34.29 ± 0.17	0.934 ± 0.006
	Federated (shared)	31.01 ± 0.09	0.941 ± 0.006
	Federated (original)	1.05 ± 0.03	0.628 ± 0.013

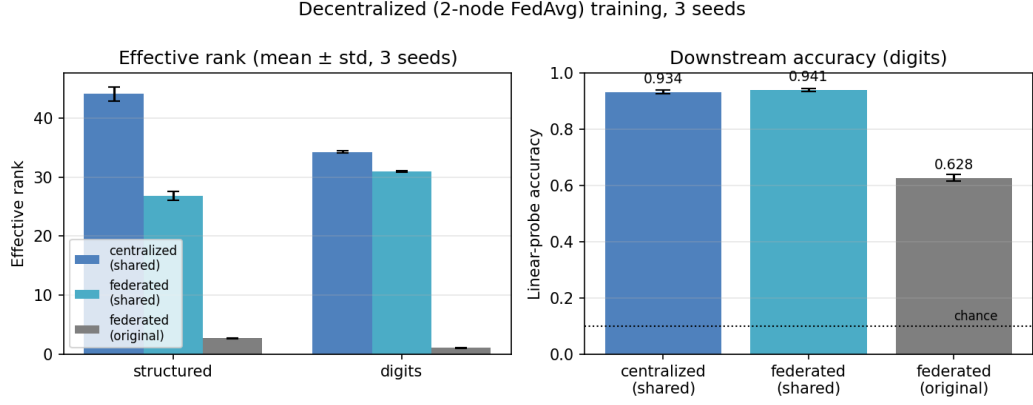


Figure 4.8: Decentralized two-node FedAvg training. With the shared-embedding placement, effective rank and downstream accuracy are preserved under federation (digits probe 0.941, matching the centralized 0.934; mean $\pm$ std over three seeds). The original wiring collapses whether trained centrally or federated.

Two conclusions follow. First, the fix is *compatible with decentralization*: federated shared-embedding training preserves high effective rank (26.89 structured, 31.01 digits) and downstream accuracy (0.941 on digits, statistically indistinguishable from the centralized 0.934), despite parameter averaging across nodes trained on disjoint data. Second, the collapse is a property of the *objective*, not the training regime: the original wiring collapses under federation exactly as it does centrally (rank 2.69/1.05, probe 0.628). Correct regularizer placement is therefore what enables collapse-resistant self-supervised learning on decentralized edge nodes—directly connecting the design principle of Section 4.5.6 to the deployment setting the architecture targets. We further stress the decentralized setting with more clients and *non-IID* data. Using Dirichlet(α) label-skewed partitions (smaller α = stronger skew) with size-weighted FedAvg, Table 4.11 and Figure 4.9 report the shared and original placements across 5–10 clients. The shared placement remains collapse-resistant throughout—effective rank 22–27 and probe accuracy 0.93–0.95—even at 10 clients with strong label skew ($\alpha = 0.1$), while the original wiring collapses to rank 1.01 and probe ≈ 0.66 in every setting. Correct placement is thus robust to the data heterogeneity that characterizes realistic federated deployment. Communication-efficiency trade-offs and heterogeneous or asynchronous clients remain for future work.

4.5.9 Scaling Up: ResNet-18 on Full CIFAR-10

The experiments so far use the lightweight CNN. To test whether the findings hold with a standard backbone at benchmark scale, we replace the encoder with a **ResNet-18** and train on all 50,000 CIFAR-10 training images, evaluating each placement with a linear probe on the 10,000-image test set (single seed, 20 epochs). Table 4.12 and Figure 4.10 report the outcome, and it *revises* the small-scale picture in two important ways.

First, the collapse is far milder with a high-capacity backbone. The original wiring reaches effective rank 9.9 rather than the ≈ 1 seen with the CNN; the catastrophic rank-1 collapse is therefore, in large part, a property of the *small model*

Table 4.11: Non-IID federated training with Dirichlet label-skewed shards (single seed). The shared placement resists collapse across client counts and skew levels; the original wiring collapses regardless.

Setting	Placement	Eff. rank	Probe
5 clients, near-IID	Shared	25.74	0.950
	Original	1.01	0.661
5 clients, moderate non-IID	Shared	27.27	0.931
	Original	1.01	0.650
10 clients, strong non-IID	Shared	21.98	0.941
	Original	1.01	0.663

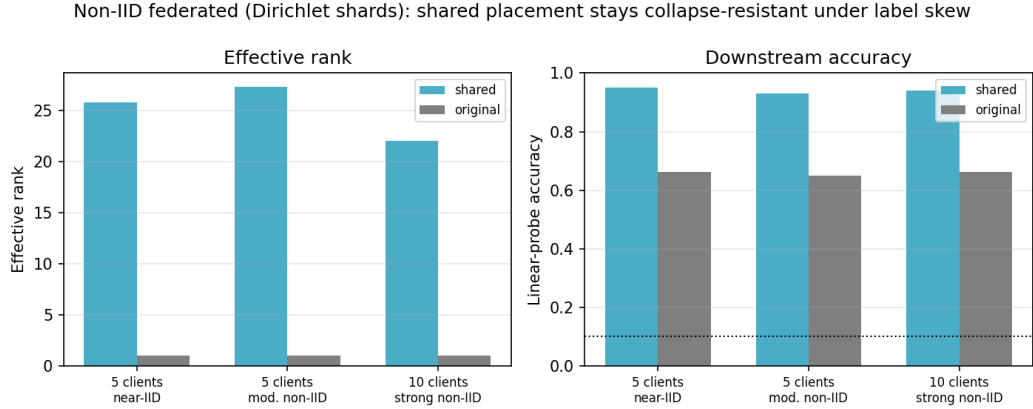


Figure 4.9: Non-IID federated training (Dirichlet shards). The shared placement preserves effective rank and downstream accuracy as the number of clients and the label skew increase; the original wiring collapses in every setting.

Table 4.12: ResNet-18 on full CIFAR-10 (single seed, 20 epochs). At this scale, effective rank no longer predicts downstream accuracy: the highest-rank configuration (shared) is the least useful, and no regularized setting beats the unregularized baseline.

Configuration	Eff. rank	Probe acc.
Unregularized	29.5	0.438
Original (var s_t , cov s_{pred})	9.9	0.421
Shared (var s_t , cov s_t)	229.2	0.358
Both s_{pred}	44.4	0.441
Swapped (var s_{pred} , cov s_t)	49.2	0.397

ResNet-18 on full CIFAR-10: high rank no longer means high accuracy (single seed, 20 epochs)

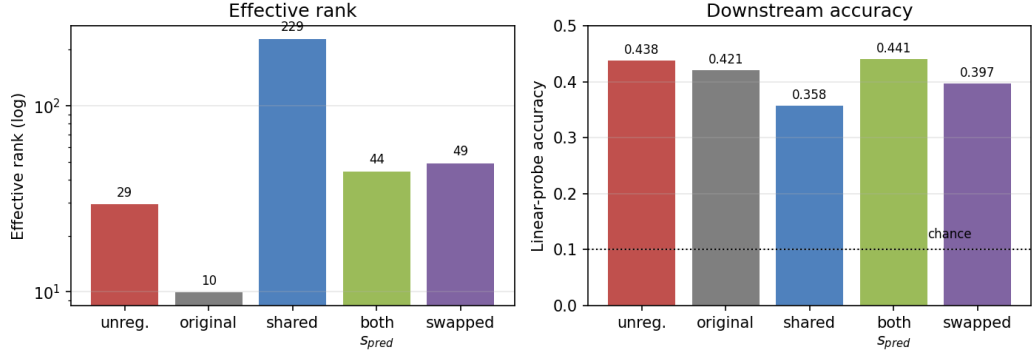


Figure 4.10: ResNet-18 on full CIFAR-10. The shared placement produces by far the highest effective rank (229) but the *lowest* downstream accuracy, and no regularized configuration beats the unregularized baseline. The rank–usefulness correspondence seen at small scale does not hold here.

rather than of the objective alone. The placement still orders the configurations—the original wiring is the lowest-rank regularized setting and the shared variant the highest (229 of 256)—so the mechanism persists, but its severity does not transfer.

Second, and more consequentially, *effective rank ceases to predict downstream usefulness at this scale*. The shared placement attains by far the highest rank yet the *lowest* probe accuracy (0.358), and no regularized configuration exceeds the unregularized baseline (0.438); the best downstream accuracy comes from the unregularized and both- s_{pred} settings (≈ 0.44). The correspondence between effective rank and linear-probe accuracy observed on the small datasets (Section 4.5.6) thus does *not* generalize to a ResNet-18 on natural images: maximizing rank via the shared reformulation does not improve—and slightly harms—the learned representation’s usefulness.

These results are reported plainly because they qualify the thesis’s central claims rather than confirm them. The placement effect on *rank* is real and reproduces across scales, but (i) its extreme form is a small-model artifact, and (ii) the link from rank to downstream quality—and hence the practical value of the shared reformulation—is scale-dependent and does not hold at ResNet/CIFAR scale. Two caveats temper even this negative result: the run is single-seed, and the self-supervised training is short (20 epochs, absolute probe accuracy ≈ 0.44), so the representations are weak in absolute terms. A longer, multi-seed study is needed to characterize the scale behavior fully; nonetheless the direction is clear enough to caution against treating effective rank as a proxy for representation quality at scale.

4.6 Discussion

On RQ1, the evidence is partial: the variance constraint demonstrably mitigates variance collapse, but at $\beta = 0.01$ the covariance constraint does not yield a high-rank representation, and the unregularized baseline is paradoxically higher-

rank—a pattern that holds on the CIFAR-10 natural-image subset as well as on the smaller datasets. A plausible explanation is that the variance and covariance penalties act on different tensors (the context embedding s_t and the prediction s_{pred} , respectively; Chapter 3) and that the covariance weight is small relative to the variance weight. The covariance-weight ablation (Section 4.5.5) tests this directly and shows that increasing β alone is insufficient to restore rank, which points to the placement of the penalties rather than their weight. The placement study (Section 4.5.6) makes this precise: collapse is unique to variance-on- s_t without a co-located covariance term, and it is invisible to both the loss and the variance criterion. Acting on that cause, the shared-embedding reformulation (Section 4.5.7) restores effective rank by more than an order of magnitude and, as the linear probe confirms, recovers downstream accuracy to the level of the unregularized baseline; Section 4.5.8 further shows the correction survives decentralized training. A crucial qualification follows from the scaling study (Section 4.5.9): the alignment between effective rank and downstream usefulness that underpins this diagnosis holds at small scale but *breaks* for a ResNet-18 on CIFAR-10, where the highest-rank configuration is the least useful and no regularizer beats the unregularized baseline. The placement effect on rank persists across scales; its practical benefit does not. On RQ2, the architecture is genuinely lightweight (2.84 MB of parameters, millisecond CPU inference), supporting its suitability for edge deployment in terms of size. On RQ3, the distillation term converges rapidly in all cases; the full objective’s total loss is non-monotonic because the covariance term grows once distillation is satisfied.

Notably, the unregularized baseline attains a far lower total loss ($\sim 10^{-3}$) precisely because, with $\alpha = \beta = 0$, the objective reduces to distillation MSE, which is trivially minimized while the representation quietly loses variance. This is a concrete illustration of why loss magnitude alone is an inadequate proxy for representation quality, and why the collapse indicators are necessary.

4.7 Threats to Validity

The following factors were examined and, where possible, addressed. Run-to-run variance, initially a concern, has been *quantified* across the study: the original collapse (five seeds, Section 4.5.4), the full placement grid (five seeds, Table 4.8), and the federated experiment (three seeds, Table 4.10) are all stable, with the key separations far exceeding their standard deviations. The main residual single-seed result is the CIFAR-10 probe. Representation *usefulness*, previously unmeasured, is now assessed by a linear probe on the labeled digits and CIFAR-10 sets (Section 4.5.7). Three factors remain open. First, scale: the study spans a lightweight CNN and a ResNet-18 on full CIFAR-10 (Section 4.5.9), but the ResNet run is single-seed and short (20 epochs), and—most importantly—it shows that the practical benefit of the reformulation does not survive at that scale, so the scale behavior is not yet settled and requires longer, multi-seed training. Second, evaluation depth: representation usefulness is measured only by a linear probe; a stronger downstream protocol may read the scaled representations differently. Third, all measurements are CPU-only, so latency and memory would differ on target embedded hardware. Addressing these is the subject of Chapter 5.

Chapter 5

Conclusion and Future Work

5.1 Summary

This thesis presented **EdgeVerify**, a Joint-Embedding Predictive Architecture designed for localized execution on edge and Dew computing nodes. The architecture combines an EMA-updated target encoder, an action-conditioned latent predictor, and a VICReg-inspired variance/covariance objective applied within a local, layer-wise distillation loss. By predicting in an abstract latent space rather than reconstructing raw inputs, and by bounding the backpropagation graph through stop-gradient firebreaks, the design targets the two coupled obstacles to on-device learning identified in Chapter 1: strict memory budgets and representational collapse.

5.2 Contributions

The work contributes: (i) the EdgeVerify architecture and its action-conditioned prediction formulation; (ii) a negative-free, stable local loss that regularizes the representation with variance and covariance terms; (iii) a proof-of-concept PyTorch implementation together with a reproducible protocol for evaluating convergence, collapse, latency, and memory footprint; (iv) a systematic regularizer-placement study yielding a design principle—applying a variance penalty to an embedding without a co-located covariance penalty on the same embedding induces rank collapse that is worse than no regularization and invisible to both the training loss and the variance criterion—validated by a linear probe; (v) a demonstration that the corrected placement’s collapse resistance transfers to decentralized federated (FedAvg) training, including under non-IID label skew with up to ten clients, whereas the flawed placement collapses in every regime; and (vi) a scaling study (ResNet-18 on full CIFAR-10) establishing that the placement effect on rank persists but is far milder with a high-capacity backbone, and that the effective-rank/usefulness correspondence—and hence the practical benefit of the reformulation—is *scale-dependent* and does not hold at ResNet/CIFAR scale.

5.3 Limitations

The principal limitation of the present work is that empirical validation is preliminary. The evaluation in Chapter 4 covered structured, digits, and a 2000-image CIFAR-10 subset, with CPU-only measurement; the structured and digits results were shown to be stable across five seeds, but a full-scale benchmark run on target hardware remains outstanding. Within those limits, the evaluation surfaced a substantive finding, reproduced across two independent environments: while the variance term prevents variance collapse, the covariance term at its current weight ($\beta = 0.01$) fails to prevent dimensional (rank) collapse, and the unregularized baseline was in fact higher-rank—a pattern that held on the CIFAR-10 natural images as well as the smaller datasets. A covariance-weight ablation (Section 4.5.5) established that this is not resolved by increasing β alone, and a shared-embedding reformulation (Section 4.5.7) then corrected it *at small scale*—restoring effective rank by more than an order of magnitude and recovering linear-probe accuracy to the unregularized baseline on the lightweight CNN. Two honest qualifications bound this result. First, even at small scale the correction removes a harmful failure mode rather than beating no-regularization. Second, and more importantly, a larger-scale study (ResNet-18 on full CIFAR-10; Section 4.5.9) shows that the extreme collapse is largely a *small-model artifact*, and that effective rank no longer predicts downstream usefulness at that scale—the shared reformulation yields the highest rank yet the lowest probe accuracy, and no regularizer beats the unregularized baseline. The placement effect on rank is robust (reproduced across five seeds and two environments and, in weakened form, at ResNet scale), but its *practical benefit is scale-dependent* and absent at ResNet/CIFAR scale. The scaling run is itself single-seed and short (20 epochs), so a longer, multi-seed study is required to settle the scale behavior. The “multi-agent” aspect is realized as federated averaging across up to ten non-IID clients, not as heterogeneous, asynchronously communicating agents.

5.4 Future Work

Several directions follow directly:

- **Explaining the scale-dependence.** The most important question raised by this work is why high effective rank fails to yield useful representations at ResNet/CIFAR scale (Section 4.5.9) when it tracks usefulness at small scale. Longer, multi-seed training at scale, a stronger self-supervised protocol, and evaluation beyond a linear probe are needed to determine whether the decoupling reflects undertraining or a genuine property of the objective in high-capacity models.
- **On-device measurement.** All timing and memory figures are CPU-only. Deploying the encoder on a defined edge device would replace them with hardware-accurate latency, memory, and energy measurements.
- **Heterogeneous and asynchronous federation.** The federated study now covers non-IID label skew and up to ten clients; realistic deployment ad-

ditionally requires heterogeneous and asynchronously communicating nodes and an analysis of communication-efficiency trade-offs [17].

- **From analysis to theory.** The linear-model account in Appendix A explains the small-scale collapse; extending it to nonlinear encoders and to a convergence (rather than fixed-point) argument would place the placement principle on a firmer footing.

Appendix A

A Linear-Model Analysis of Placement-Induced Collapse

This appendix is a draft analysis to be verified and adapted. It gives a first-principles explanation, in a linearized model, of why applying the variance penalty to the context embedding s_t without a co-located covariance penalty on s_t induces rank collapse—and why that collapse is invisible to both the training loss and the variance criterion. The argument reproduces the three empirical signatures reported in Section 4.5.6: a final variance loss of 0, the largest $\bar{\sigma}$, and effective rank ≈ 1 .

A.1 Setup

Approximate the context encoder and the predictor by linear maps. Let the input be $x \in \mathbb{R}^m$ with $x \sim \mathcal{N}(0, I_m)$, the context embedding be

$$s_t = Wx, \quad W \in \mathbb{R}^{d \times m}, \quad (\text{A.1})$$

and the predictor be a trainable linear map $s_{\text{pred}} = P s_t$ with $P \in \mathbb{R}^{d \times d}$ (absorbing the action conditioning into P for this analysis). Write the context-embedding covariance as

$$\Sigma = \text{Cov}(s_t) = WW^\top \in \mathbb{R}^{d \times d}. \quad (\text{A.2})$$

The two VICReg-style penalties are, for a chosen tensor u with covariance Σ_u ,

$$\mathcal{L}_{\text{var}}(u) = \frac{1}{d} \sum_{j=1}^d \max(0, \gamma - \sqrt{(\Sigma_u)_{jj}}), \quad (\text{A.3})$$

$$\mathcal{L}_{\text{cov}}(u) = \frac{1}{d} \sum_{i \neq j} (\Sigma_u)_{ij}^2. \quad (\text{A.4})$$

The variance penalty constrains only the *diagonal* of Σ_u ; the covariance penalty constrains only the *off-diagonal*.

A.2 Original wiring: variance on s_t , covariance on s_{pred}

The objective is $\mathcal{L} = \mathcal{L}_{\text{distill}} + \alpha \mathcal{L}_{\text{var}}(s_t) + \beta \mathcal{L}_{\text{cov}}(s_{\text{pred}})$.

The covariance term imposes no constraint on the rank of Σ . The covariance of the prediction is $\Sigma_{\text{pred}} = P\Sigma P^\top$. For *any* Σ with eigendecomposition $\Sigma = U\Lambda U^\top$, choosing $P = DU^\top$ for any diagonal D gives $\Sigma_{\text{pred}} = D\Lambda D$, which is diagonal, so $\mathcal{L}_{\text{cov}}(s_{\text{pred}}) = 0$. The predictor can therefore *absorb* the decorrelation for any context covariance, including a rank-deficient one. Consequently $\beta \mathcal{L}_{\text{cov}}(s_{\text{pred}})$ places no constraint on $\text{rank}(\Sigma)$.

The variance term admits rank-1 solutions. The only remaining constraint on s_t is $\mathcal{L}_{\text{var}}(s_t) = 0 \iff \Sigma_{jj} \geq \gamma^2 \forall j$. Consider a rank-1 covariance $\Sigma = vv^\top$ with $v \in \mathbb{R}^d$ and $|v_j| \geq \gamma$ for all j . Then $\Sigma_{jj} = v_j^2 \geq \gamma^2$, so $\mathcal{L}_{\text{var}}(s_t) = 0$ and the per-dimension standard deviation is $\sqrt{\Sigma_{jj}} = |v_j| \geq \gamma$, giving $\bar{\sigma} \geq \gamma$. Its effective rank is exactly 1.

Conclusion. A rank-1 context representation with inflated per-dimension scale is a global optimum of the original-wiring objective: it drives $\mathcal{L}_{\text{var}}$ and $\mathcal{L}_{\text{cov}}$ to zero and (with a suitable predictor) matches the distillation target. At this optimum the variance loss is exactly 0, the mean standard deviation $\bar{\sigma}$ is large, and the effective rank is 1—precisely the empirical signature of Table 4.7 (final variance loss 0.000, largest $\bar{\sigma}$, rank ≈ 1). Because both the loss and the variance criterion are satisfied, the collapse is invisible to them.

A.3 Shared placement: both penalties on s_t

Now the objective includes $\beta \mathcal{L}_{\text{cov}}(s_t) = \frac{\beta}{d} \sum_{i \neq j} \Sigma_{ij}^2$, acting on the same covariance Σ that the variance term constrains. A rank-1 $\Sigma = vv^\top$ now incurs an off-diagonal penalty

$$\sum_{i \neq j} (v_i v_j)^2 = \left(\sum_j v_j^2 \right)^2 - \sum_j v_j^4, \quad (\text{A.5})$$

which is large whenever the diagonal constraint $|v_j| \geq \gamma$ holds. Collapse is therefore no longer optimal. The joint stationary condition drives the off-diagonals of Σ toward 0 while the diagonals are held at γ^2 , i.e. $\Sigma \rightarrow \gamma^2 I_d$, which is full rank d . This is the standard VICReg fixed point, and it explains why co-locating the covariance term on s_t restores effective rank (Table 4.7, shared cells).

A.4 Why co-location is necessary

The asymmetry is structural. The variance penalty is a per-coordinate (diagonal) constraint and is blind to inter-dimensional correlation. Rank deficiency is an off-diagonal (correlation) property. Preventing it therefore requires imposing the decorrelation penalty on the *same* representation whose rank is at stake. Routing that penalty through the trainable predictor ($s_{\text{pred}} = Ps_t$) lets P satisfy it for any Σ , decoupling it from the rank of s_t . Hence a variance penalty without a co-located covariance penalty inflates the diagonal while leaving the rank unconstrained, and the collapsed optimum is invisible to the objective—exactly the phenomenon characterized empirically in this thesis.

A.5 Limitations of the analysis

The argument treats the encoder and predictor as linear and the input as isotropic Gaussian; it characterizes the optimality of collapsed *fixed points* rather than proving that gradient descent converges to them, and it abstracts away batch normalization, the ReLU nonlinearity, and the EMA target dynamics. A full nonlinear treatment, and a convergence (rather than optimality) analysis, are left as future work. Nonetheless, the linear model already predicts the three measured signatures of the collapse and the effect of co-locating the penalties, which is the intended purpose of this appendix.

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